Data-Driven Optimization for Operational Efficiency at Meridian Restaurant, Tirupati

A proposal report for the BDM capstone Project

Submitted by

Name: Rudra Rajeswari

Roll number: 24f1001011

Email: 24f1001011@ds.study.iitm.ac.in

IITM Online BS Degree Program,

Indian Institute of Technology, Madras, Chennai

Tamil Nadu, India, 600036

Contents

1 Executive Summary and Title…….4

2 Organisation Background ……...4

3 Problem Statement …….5

3.1 Problem statement 1……...5

3.2 Problem Statement 2 …….5

3.3 Problem Statement 3……..5

4 Background of the Problem……....5

5 Problem solving Approach ……....6

6 Expected Timeline …….7

7 Expected Outcome …….8

Declaration Statement

I am working on a Project Titled "Data-Driven Optimization for Operational Efficiency at Meridian Restaurant, Tirupati **".** I extend my appreciation to Meridian, for providing the necessary resources that enabled me to conduct my project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered through primary sources and carefully analyzed to assure its reliability.

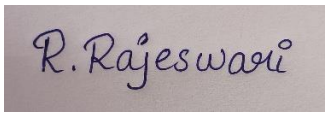
Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the information of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I agree that all the recommendations are business-specific and limited to this project exclusively, and cannot be utilized for any other purpose with an IIT Madras tag. I understand that IIT Madras does not endorse this.

Signature of the candidate:



Name : Rudra Rajeswari

Date: 13-10-2025

Data-Driven Optimization for Operational Efficiency at Meridian Restaurant, Tirupati

1.Executive Summary

The Meridian Restaurant is a restaurant that opened in 2020 by Mr. Gopi Krishna located in Vidyanagar, Kota, Tirupati District, Andhra Pradesh and works between the time of 11:30 AM to 9:30 PM and has gained its reputation in the region because of the quality food and service provided. Meridian Restaurant is a B2C business where it deals directly with the individual customers by providing them with freshly cooked food. Though successful, the small-scale restaurant has operational setbacks typical of semi-rural businesses, including record-keeping that is not computerized and less use of data.

Some of the main obstacles are wastage of ingredients through lack of stock tracking and irregularities in the staff that impact day-to-day activities. To address these problems, the restaurant will use a data-driven strategy aiming at three main sections: inventory management, business performance analysis, and workforce planning.

Through adoption of such technologies as predictive inventory and stocking, sales trend and attendance tracking, Meridian will be able to minimize wastage, improve staff scheduling, and have a better comprehension of the customer preferences. These information-driven plans will be useful in making operations more streamlined, efficient, and allow the company to continue growing steadily in a localized market that grows increasingly competitive.

2. Organization Background

Meridian Restaurant is a small, owner-managing eatery that was established in 2020 by Mr. Gopi Krishna in Vidyanagar, Kota, Tirupati District, Andhra Pradesh,opposite S.I.T. Balabarathi School and below the Gym Building. It functions between the hours of 11: 30 AM to 9: 30 PM and provides fresh vegetarian and non-vegetarian cuisine at low prices to the local residents, students, and families. The restaurant now has about 10-12 employees who work on rotating bases, that is, cookers, servers, and assistants.

Meridian continues to be dependent on manual operations despite its consistent growth. Every sale, cost and record in transactions are kept in hand written notebooks which result in inconsistency of data and complicate tracking of inventory or estimating profits. Lack of digital tools restricts efficiency and proper cost analysis and decision-making.

3. Problem Statement

3.1 Perk up Workforce and Resource Planning:

- Employee attendance, turnover of employees and trends in their performance to maximize shift.
- Plan hiring and training using facts and be sure that there is sufficient staffing at peak times.

3.2 Maximize Inventory and Ingredient Control:

- Install actual time monitoring of ingredient consumption to minimize overstocking and spoilage.
- Predictive analytics can be used to predict future inventory levels using the historical consumption pattern.

3.3 Business performance data analytics:

It involves examining business performance with data analytics.

- Analyze the sales statistics to know the busiest hours and the best menu products to allocate more resources towards them.
- Examine customer preferences, customer feedbacks to enhance customer satisfaction and improve menu offerings.

4.Background of the Problem

Meridian Restaurant is a restaurant that has a strong foundation and has been gradually expanding because of its quality and cheap food. It was established by Mr. Gopi Krishna in 2020 in Vidyanagar. But lack of an appropriate data management system has started to hit operations and profits. The restaurant is still relying on manual entry of the sales, expenses, and inventory to keep records hence it is hard to trace the stock and trace the leakage of finances. There has also been inconsistency in service and wastage of ingredients due to frequent staff turnover and poor utilization of inventory thus reducing profit margins. Without a combined system of sales, purchases and expenses, the owner will be unable to determine the profitability and predict demand. To be sustainable over the long-term, the restaurant should consider a digital approach to data management that involves automated tracking, real-time insights, and analytics to determine trends, use resources to their maximum capacity, and improve the overall efficiency of the restaurant.

5. Problem Solving approach

5.1 Methods:

To overcome operational inefficiencies, a data-driven analytical method which is a combination of descriptive, diagnostic, and predictive analytics will be implemented. Descriptive analytics will help summarize the current sales, costs, and stock quantities; diagnostic analytics will help identify the causes of wastage, unstable sales, and employee turnover; and predictive analytics will help predict the sales trends and demand in the future and staffing requirements. The multi-step model allows making decisions supported by data that can enhance efficiency in the short term and promote sustainability in the long term.

5.2 Data Collection:

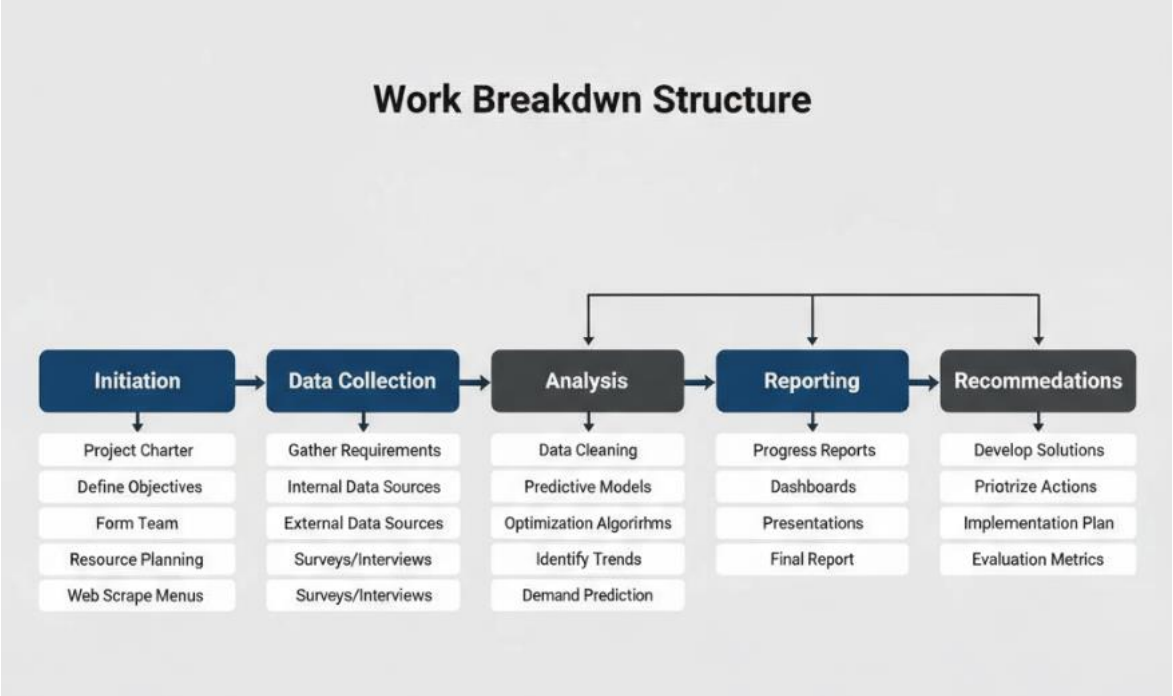
Primary and secondary data will be used. The primary data collection will be obtained by discussing with the owners and conducting observations on the site, whereas secondary data will be digitized by using available records. The important databases are sales (date, amount, items, mode of payment), expenses (purchases, salaries, utilities), inventory (stock level, usage, wastage), and employee (attendance, tenure). These transparency, resource optimization and integrated management support goals.

5.3 Analysis Tools:

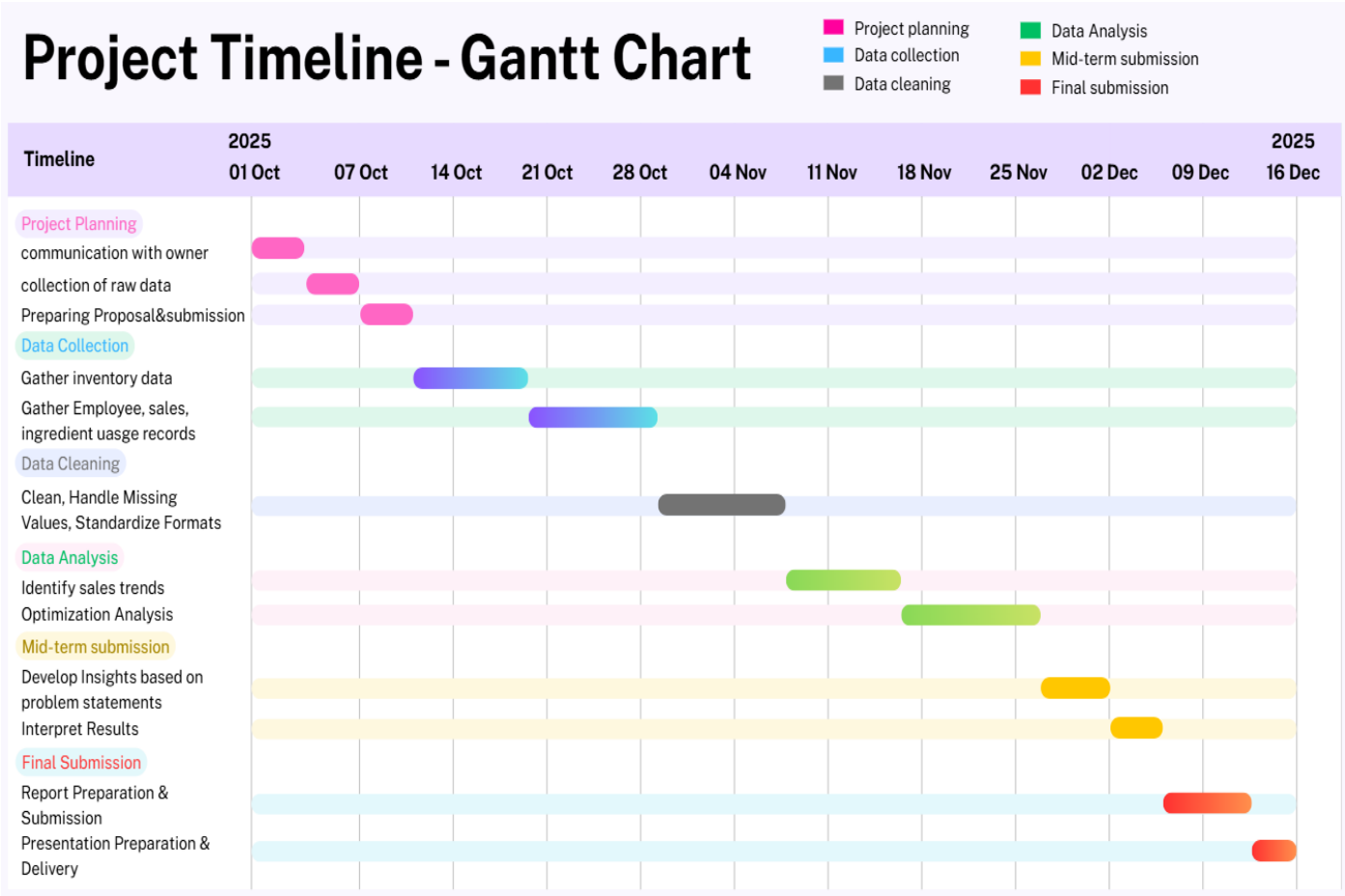
Analysis will be done using Python and Microsoft Excel. By using Python (along with Pandas, Matplotlib, and Jupyter Notebook), statistical analysis, visualization, and automation are possible, whereas in Excel, the correct entry of data and its validation are guaranteed. This is a scalable, low-cost toolset, which enables informed decisions and an easy change between manual and digital management.

6. Expected Timeline

6.1 Work Breakdown Structure (WBS):



6.2 Gantt chart:



7. Expected Outcome

The project is expected to give Meridian Restaurant practical and data-driven key insights to enhance efficiency and decision-making. Through business data management, the restaurant will be able to convert raw information into workable strategies in inventory, operations and workforce management.

Regarding inventory control, information analysis will assist in monitoring the consumption or use of ingredients, avoid overstocking, and minimize wastage. This will also facilitate proper record-keeping and make smarter purchase decisions which will make operations cost-effective.

To the business performance, the past sales data and customer data will be analyzed to determine the peak business hours, menu items that are more demanded, and the preferences of the customers. The insights will be used in improving the resources allocation, menu as well as customer satisfaction.

Workforce and resource planning will be the main focus of the project which will solve the problems of irregularity and turnover of staff members by researching the attendance, performance, and scheduling patterns. This will help in better staffing predictions, better retention techniques and service delivery.

The presentation of all the findings will be in the form of visual dashboard and reports that will give a picture of inventory level, sales and workforce measures. The outcomes will transform the restaurant to data conversion based decision-making, better cost management, greater productivity, and long-term development.